



HET-CD-004

(Thermocouple Subnode)

Author: Electrical & Software Team / Coren Benavides / Max Day / Matthew Fan

Section 1 Component Scope

A brief introduction of the thermocouple is outlined in this section.

1.1 Overview

The thermocouple functions as a temperature sensor that can read extremely high temperatures through physical contact. Its intended application is to read temperature to study temperature effects on the hall thruster system. As shown in Fig. 1 below, the thermocouple consists of two wire leads from different metals. On one end, the two metal wire leads are joined at a junction to create a thermocouple in which to sense temperature. On the other end, these metal wires are joined together by an isothermal cold junction that initially holds both thermocouple leads at the same temperature. When the thermocouple is in physical contact with the thermal target, the temperature from the thermocouple reaches the junction block. When a metal bar (or block in this case) is heated on one end, it creates a voltage (known as the Seebeck voltage) that spans across the length of the bar. Since the two metals have dissimilar thermal properties and thus dissimilar Seebeck voltages, a thermocouple voltage V_{TC} is generated. From this voltage, the temperature can be measured from the thermal target.

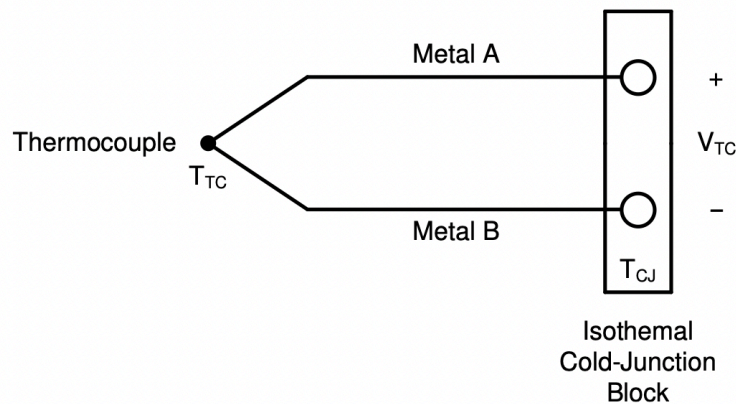


Figure 1: Thermocouple Schematic

Several thermocouples will be placed strategically in the hall thruster system to study temperature effects of various components in the system such as the cathode, anode, and electromagnetic coils. For the cathode, reducing the temperature gradient can increase the life time of the cathode, allowing it to be used for multiple runs or longer periods of time. For the anode, a high temperature could cause erosion to the anode which leads to a decrease in electric potential which thus leads to a decrease in thrust. For the electromagnetic coils, increasing the magnet temperature decreases the magnetic strength and thus decreases the thrust efficiency.

1.2 Role in Assembly

As previously mentioned, there will be multiple thermocouples needed to measure temperature at different locations in the hall thruster system. The identified locations, shared by the thermal team, are presented in Table 1 below.

Table 1: Thermocouple Locations

Thermocouple No.	Location(Expected max temperature)
1	Cathode(800C)
2	Electromagnetic Coil(250C)
3	Front Plate(400C)
4	Back Plate/Heat Sink(250C)

All thermocouples will be connected with an ESP32 to relay temperature readings.

1.3 Design Objectives

Table 2 outlines the key design goals for the thermocouple.

Table 2: Key Design Goals

1. The thermocouples shall be rated to measure 800C on the cathode and 600C for every other component of the hall thruster.
2. The thermocouple shall be able to withstand high temperature and heat fluxes.
3. The thermocouple shall have a sufficient mass and volume.
4. The thermocouple shall interface with all necessary components.

Section 2 Requirements and Constraints

Requirements and constraints are outlined in this section. Unless specified, all requirements and constraints must be met for the

2.1 Performance Requirements

Table 3 outlines the performance requirements for the thermocouple to successfully function.

Table 3: Key Performance Requirements

1. The thermocouple shall accurately measure the temperature of the hall thruster system.
2. The thermocouple shall maintain stable readings of temperature inside the vacuum chamber.
3. The thermocouple shall maintain durability throughout the hot fire test.
4. The thermocouple shall properly transmit data to the ESP32 board for data processing.
5. The thermocouple shall be operational throughout the hot fire test.

2.2 Physical Constraints

The thermocouple must have a sufficient mass and volume to operate properly.

2.3 Manufacturing Constraints

There are no manufacturing constraints, because there is no need for manufacturing any parts of the thermocouple. All parts will be purchased from outside sources. Section 4 outlines the plan for manufacturing. The section will be left blank.

2.4 Environmental Constraints

The thermocouple must be operational under vacuum and high-temperature conditions.

2.5 Budgetary Constraints

The total budget of all thermocouples must not exceed 225 dollars. This budget is determined due to the need to purchase multiple thermocouples.

Section 3 System Design

3.1 Component Selection

The selected component for the thermocouple is the K-type thermocouple (see Fig. 2 below). The component is able to measure up to 2300 Fahrenheit or 1260 Celsius, which is sufficient since the hull thruster is expected to have a maximum temperature reading of 1100 Celsius.



Figure 2: Example of a K-type Thermocouple

A second component is needed for the ESP32 to read the thermocouple through a junction. A MAX6675 amplifier (see Fig. 3 below) is able to convert the voltage reading into a temperature reading and relay the data to the ESP32. The MAX6675 amplifier was selected, because it is not only compatible with a K-type thermocouple but can be purchased with one as well. Since only one MAX6675 amplifier can pair with a K-type thermocouple, purchasing the complete set is cheaper than purchasing them separately.

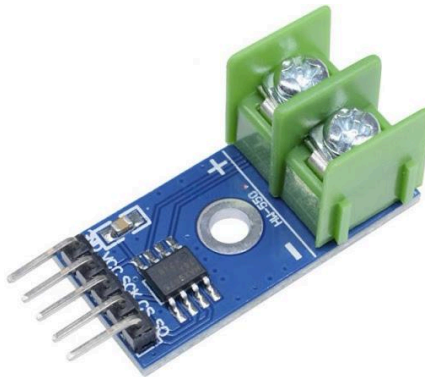


Figure 3: MAX6675 Amplifier with K-type Thermocouple

Thermocouple Samples:

600 - 800 °C - K-Type

Thermocouple K-Type Temperature Range 0-800C Sensor Probe - LOW END \$14.10	https://www.twinschip.com/Thermocouple-K-Type-Temperature-Range-0-800C-Sensor-Probe	Dimensions: 100.00cm x 1.00cm x 1.00cm, weight 40 grams, k-type. Temperature range 0-800°C with metal shielding and internal fiberglass insulation.
Inconel Thermocouple Probe	https://evosensors.com/colle	Temperatures range from

Type K 1/8" Diameter 12 Inches Long Ungrounded with Miniature Male Connector. - MID RANGE \$47.00	https://www.grainger.com/product/inconel-600/products/inconel-sheath-type-k-thermocouple-probe-1-8-diameter-12-inches-long-ungrounded-with-miniature-male-connector	-200 to 1070 °C. "Excellent temperature solution for engine technology".
TEMPCO Thermocouple Probe: 32°F to 2,300°F /0°C to 1,260°C, +/-2.2°C or +/-0.75% Accuracy, Type K - LOW END \$23.75	https://www.grainger.com/product/3AAA2?gucid=N:N:FPL:Free:GGL:CSM-1946:tew63h3:20501231	Temperature ranges from 0-1260°C. Inexpensive with high temperature range.
TJ36-CAIN-14U-6-CC-XCIB HIGH END \$153.59	https://www.dwyeromega.com/en-us/rugged-thermocouple-transition-joint-probes-with-nextel-insulation/TJ36-XCIB/p/TJ36-CAIN-14U-6-CC-XCIB	0-920°C temperature range. Very expensive but very high quality. Likely the best option for the 800°C measurement. Very customizable but might have stocking issues.
Digi-Sense Inconel Type K Thermocouple Probe, Quick Dis-connector, 6" L x 1/16" Dia, Grounded MID RANGE \$49.80	https://www.coleparmer.com/i/digi-sense-inconel-type-k-thermocouple-probe-quick-dis-connector-6-l-x-1-16-dia-grounded/1852359?PubID=UX&persist=true&ip=no	-150° to 865°C range. Relatively inexpensive but might take a while to ship.

3.2 Circuit Schematic

A circuit schematic is shown in Fig. 4 below to outline the appropriate connections between the ESP32, and MAX6675 amplifiers. It is important to note that the MAX6675 amplifiers connect to the K-type thermocouple wire leads.

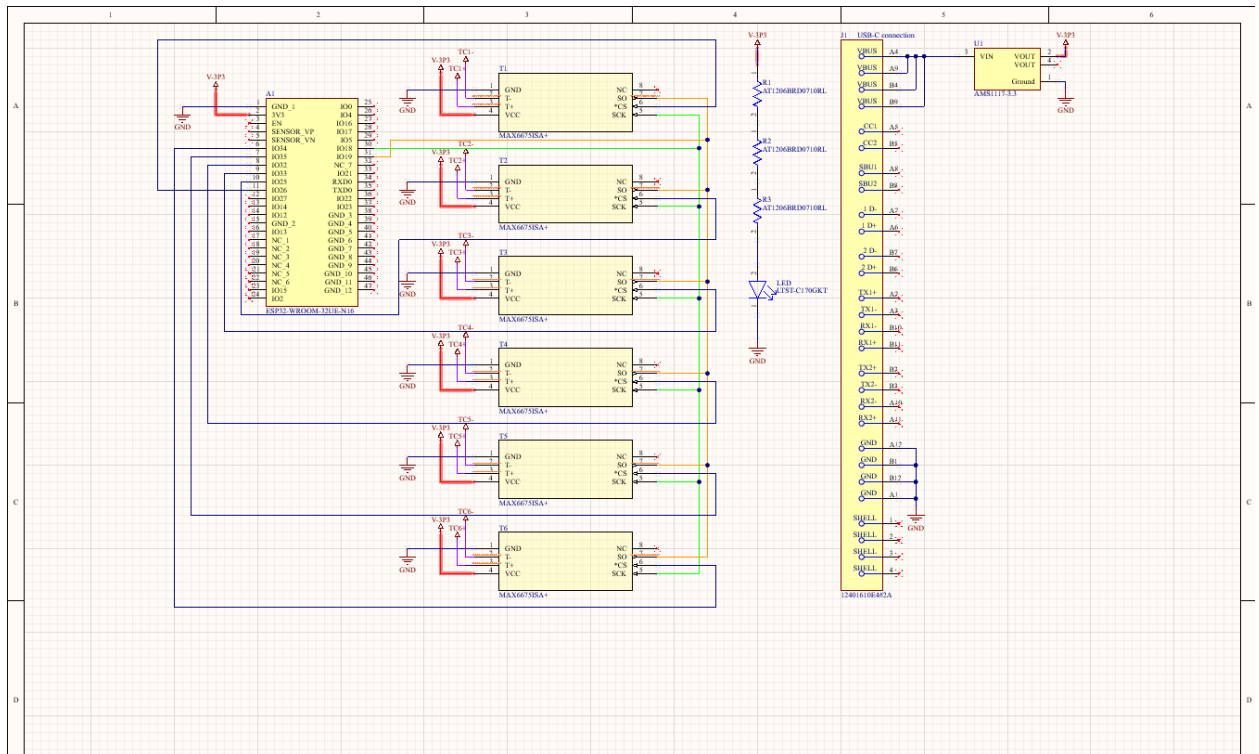


Figure 4: Circuit Schematic

3.3 Interface Definition

All components of the circuit schematic (see Fig. 5 above) need to interface properly. The necessary pins and interface connections for each component. Mechanically the thermocouples will be attached to the hall effect thruster via epoxy or kapton tape. The sensor board will have the MAX6675 amplifiers on them and will be housed away from the thruster itself. Depending on where the housing will be mounted it can be fastened to the test stand by an adhesive, velcro, or screws. This will allow all of the thermocouple electronics to be confined into a single space.

3.4 Analysis and Validation

Section 4 Manufacturing Plan

The MAX6675 amps will be soldered to the sensor board and the thermocouples themselves will attach to screw terminals on the sensor board.

Section 5 Testing and Verification

The thermocouple needs to be tested and verified prior to the final test.

5.1 Pre-Test Requirements

- Make sure that all wires are properly secured and have the right polarity.
- Ensure that all pins are connected as shown in the wire schematic.

5.1.1 Test Setup

Two to six thermocouples will be interfaced with an MCU. Precreated test code will be used to initialize, calibrate, and then test the thermocouples. For the test we will need the following items, a calibrated thermometer, a cup of ice water, and heat pack/pad).

5.2 Verification Tests

The thermocouples will then be attached to areas at varying known temperatures (ambient air, ice cold water, heat pack/pad). Each thermocouple will be dipped in the ice cup, wait for temperature to stabilize, then it will be taken out, let temperature stabilize, and then it will touch the heat pack/pad, and let the temperature stabilize. This test will verify that the calibration of the thermocouples was successful, it will show how heat is dissipated from the thermocouple (we can see the “rise time”), and ensure that the temperature can be accurately read across a temperature range. This test will also ensure that the code is fully operational.

Section 6 Safety and Risk Mitigation

The risks of this component are limited to only electrical.

6.1 Safety & Risk Mitigation

List hazards (sharp edges, electrical exposure, chemical risks, etc.).

After a test the thermo couples will be hot and will need a little time to cool down before handling. There will be no conductive liquids allowed to be uncapped near these components in order to prevent any spills causing short circuiting. There are no other known risks of the thermocouple subnode.

Section 7 Cost Analysis

Depending on the temperatures and accuracy required the thermocouple in question can be relatively cheap or expensive. For our mid to high temperatures the price of the thermo couple is around \$50. Our low thermocouple is about \$14.

7.1 Bill of Materials

Quantity: 1	Digi-Sense Inconel Type K Thermocouple Probe, Quick Dis-connector, 6" L x 1/16" Dia, Grounded MID RANGE \$49.80	https://www.coleparmer.com/i/digi-sense-inconel-type-k-thermocouple-probe-quick-dis-connector-6-l-x-1-16-dia-grounded/1852359?PublD=UX&persist=true&ip=no	-150° to 865°C range. Relatively inexpensive but might take a while to ship.
Quantity: 3	Thermocouple K-Type Temperature Range 0-800C Sensor Probe - LOW END \$14.10	https://www.twinschip.com/Thermocouple-K-Type-Temperature-Range-0-800C-Sensor-Probe	Dimensions: 100.00cm x 1.00cm x 1.00cm, weight 40 grams, k-type. Temperature range 0-800°C with metal shielding and internal fiberglass insulation.

600 - 800 °C - K-Type

Thermocouple K-Type Temperature Range 0-800C Sensor Probe - LOW END \$14.10	https://www.twinschip.com/Thermocouple-K-Type-Temperature-Range-0-800C-Sensor-Probe	Dimensions: 100.00cm x 1.00cm x 1.00cm, weight 40 grams, k-type. Temperature range 0-800°C with metal shielding and internal fiberglass insulation.
Inconel Thermocouple Probe Type K 1/8" Diameter 12 Inches Long Ungrounded with Miniature Male Connector. - MID RANGE \$47.00	https://evosensors.com/collections/inconel-600/products/inconel-sheath-type-k-thermocouple-probe-1-8-diameter-12-inches-long-ungrounded-with-miniature-male-connector	Temperatures range from -200 to 1070 °C. "Excellent temperature solution for engine technology".
TEMPCO Thermocouple Probe: 32°F to 2,300°F /0°C to 1,260°C, +/-2.2°C or +/-0.75% Accuracy, Type K - LOW END \$23.75	https://www.grainger.com/product/3AAA2?gucid=N:N:FPL:Free:GGL:CSM-1946:tew63h3:20501231	Temperature ranges from 0-1260°C. Inexpensive with high temperature range.
TJ36-CAIN-14U-6-CC-XCIB HIGH END \$153.59	https://www.dwyeromega.com/en-us/rugged-thermocouple-transition-joint-probes-with-nextel-insulation/TJ36-XCIB/p/TJ36-CAIN-14U-6-CC-XCIB	0-920°C temperature range. Very expensive but very high quality. Likely the best option for the 800°C measurement. Very customizable but might have stocking issues.
Digi-Sense Inconel Type K Thermocouple Probe, Quick Dis-connector, 6" L x 1/16" Dia, Grounded MID RANGE \$49.80	https://www.coleparmer.com/i/digi-sense-inconel-type-k-thermocouple-probe-quick-dis-connector-6-l-x-1-16-dia-grounded/1852359?PubID=UX&persist=true&ip=no	-150° to 865°C range. Relatively inexpensive but might take a while to ship.

Section 8 References

Document No.	Document Title	Date	Author
1	A Basic Guide to Thermocouple Measurements	March 2023	Joseph Wu
2	Thermal Characterization of a Hall Effect Thruster	March 2008	Alex M. Bohnert
3	Cold-Junction-Compensated K-Thermocouple-to-Digital Converter	June 2021	Maxim Integrated Products
4	8-Bit I/O Expander with Serial Interface	March 2019	Microchip Technology Inc.

Section 9 Appendices

Appendix A –